Newman-Penrose Formalism Calculations

Stephani et al., Equation (12.24a)

Metric

$$ds^2 = -\mathrm{d}t \otimes \mathrm{d}t + \left(\frac{16\,a^4}{16\,a^4 + 8\,a^2 r^2 + r^4}\right) \mathrm{d}r \otimes \mathrm{d}r + \left(\frac{16\,a^4 r^2}{16\,a^4 + 8\,a^2 r^2 + r^4}\right) \mathrm{d}\theta \otimes \mathrm{d}\theta + \left(\frac{16\,a^4 r^2 \sin(\theta)^2}{16\,a^4 + 8\,a^2 r^2 + r^4}\right) \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{2\sqrt{2}a^2}{4\,a^2 + r^2}\right) dr$$

$$n = \left(\frac{1}{2}\sqrt{2}\right) dt + \left(-\frac{2\sqrt{2}a^2}{4\,a^2 + r^2}\right) dr$$

$$m = \left(\frac{2\sqrt{2}a^2 r}{4\,a^2 + r^2}\right) d\theta + \left(\frac{2i\sqrt{2}a^2 r \sin(\theta)}{4\,a^2 + r^2}\right) d\phi$$

$$\overline{m} = \left(\frac{2\sqrt{2}a^2 r}{4\,a^2 + r^2}\right) d\theta + \left(-\frac{2i\sqrt{2}a^2 r \sin(\theta)}{4\,a^2 + r^2}\right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{\sqrt{2}(4\,a^2 + r^2)}{8\,a^2}\right) dr$$

$$n = \left(-\frac{1}{2}\sqrt{2}\right)\, dt + \left(-\frac{\sqrt{2}\left(4\,a^2 + r^2\right)}{8\,a^2}\right)\, dr$$

$$m = \left(\frac{\sqrt{2}\left(4\,a^2 + r^2\right)}{8\,a^2 r}\right)\, d\theta + \left(\frac{\sqrt{2}\left(4i\,a^2 + i\,r^2\right)}{8\,a^2 r \sin\left(\theta\right)}\right)\, d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}\left(4\,a^2 + r^2\right)}{8\,a^2 r}\right)\, d\theta + \left(-\frac{\sqrt{2}\left(4i\,a^2 + i\,r^2\right)}{8\,a^2 r \sin\left(\theta\right)}\right)\, d\phi$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\,\partial_t + \frac{\sqrt{2}\left(4\,a^2 + r^2\right)}{8\,a^2}\,\partial_r$$

$$\Delta = -\frac{1}{2}\sqrt{2}\,\partial_t - \frac{\sqrt{2}\left(4\,a^2 + r^2\right)}{8\,a^2}\,\partial_r$$

$$\delta = \frac{\sqrt{2}\left(4\,a^2 + r^2\right)}{8\,a^2 r}\,\partial_\theta + \frac{\sqrt{2}\left(4i\,a^2 + i\,r^2\right)}{8\,a^2 r \sin\left(\theta\right)}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}\left(4\,a^2 + r^2\right)}{8\,a^2 r}\,\partial_\theta - \frac{\sqrt{2}\left(4i\,a^2 + i\,r^2\right)}{8\,a^2 r \sin\left(\theta\right)}\,\partial_\phi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}(2\,a + r)(2\,a - r)}{8\,a^2 r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}(2\,a + r)(2\,a - r)}{8\,a^2 r}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\left(4\,a^2 + r^2\right)\cos\left(\theta\right)}{16\,a^2 r\sin\left(\theta\right)}$$

$$\beta = \frac{\sqrt{2}\left(4\,a^2 + r^2\right)\cos\left(\theta\right)}{16\,a^2 r\sin\left(\theta\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = \frac{1}{2\,a^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{4\,a^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{1}{2\,a^2}$$

$$\Lambda = \frac{1}{4\,a^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"